Readme

1. This Appendix presents other analytical expressions for gravitational vector (VxLS and VyLS), gravitational gradient tensor (VxyLS, VxzLS, and VyzLS), and gravitational curvature (VxyzLS, VzzxLS, and VzzyLS).
2. The analytical expressions for gravitational curvature (VxxxLS, VxxyLS, VyyxLS, and VyyyLS) are not presented because the situation $\frac{1}{0}$ occurs.
3. EllipticE gives the complete elliptic integral $E(m) = E\left(\frac{\pi}{2} \middle| m\right)$. For $-\pi/2 < \phi < \pi/2$ and $m < 1$, $E(\phi \mid m) = \int_0^{\phi} \left(1 - m \sin^2(\theta)\right)^{1/2} d\theta$.

EllipticK gives the complete elliptic integral of the first kind $K(m) = F\left(\frac{\pi}{2} \middle| m\right)$. For real m, $-\pi/2 < \phi < \pi/2$ and $m \sin^2(\phi) < 1$, $F(\phi \mid m) = \int_0^{\phi} \left(1 - m \sin^2(\theta)\right)^{-1/2} d\theta$.

For certain special arguments, EllipticE and EllipticK automatically evaluate to exact values and can be evaluated to arbitrary numerical precision.

1. Other analytical expressions for gravitational vector, gravitational gradient tensor, and gravitational curvature

$$\text{VxLS} = \frac{1}{3\,r^2}\,G\rho\left((r-r1)\,\left(2\,r^2-r1^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]-(r-r2)\,\left(2\,r^2-r2^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]-(r+r1)\,\left(2\,r^2+r1^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]+(r+r2)\,\left(2\,r^2+r2^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]\right)\,(\text{Sin}[\lambda-\lambda1]-\text{Sin}[\lambda-\lambda2])$$

$$\text{VyLS} = \frac{1}{3\,r^2}\,G\rho\,(\text{Cos}[\lambda-\lambda1]-\text{Cos}[\lambda-\lambda2])\left(-\left((r-r1)\,\left(2\,r^2-r1^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]\right)+(r-r2)\,\left(2\,r^2-r2^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]+(r+r1)\,\left(2\,r^2+r1^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]-(r+r2)\,\left(2\,r^2+r2^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]\right)$$

$$\text{VxyLS} = \frac{1}{3\,r^3}\,G\rho\left(-2\,(r1-r2)\,\left(3\,r^2+r1^2+r1\,r2+r2^2\right)+3\,r^3\left(\text{Log}\left[\frac{r-r2}{r-r1}\right]+\text{Log}\left[\frac{r+r1}{r+r2}\right]\right)\right)\,\text{Sin}[2\,\lambda-\lambda1-\lambda2]\,\text{Sin}[\lambda1-\lambda2]$$

$$\text{VxzLS} = \frac{1}{3\,r^3}\,4\,G\,\rho\,\text{Cos}\left[\lambda - \frac{\lambda 1}{2} - \frac{\lambda 2}{2}\right]\left(-\left((r-r1)\,\left(r^2+r1^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]\right) + (r-r2)\,\left(r^2+r2^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right] + \frac{\left(r^4+r^2\,r1^2+r1^4\right)\,\text{EllipticK}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]}{r-r1} - \frac{\left(r^4+r^2\,r2^2+r2^4\right)\,\text{EllipticK}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]}{r-r2}\right)\text{Sin}\left[\frac{\lambda 1-\lambda 2}{2}\right]$$

$$\text{VyzLS} = \frac{1}{3\,r^3}\,4\,G\,\rho\left(-\left((r-r1)\,\left(r^2+r1^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]\right) + (r-r2)\,\left(r^2+r2^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right] + \frac{\left(r^4+r^2\,r1^2+r1^4\right)\,\text{EllipticK}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]}{r-r1} - \frac{\left(r^4+r^2\,r2^2+r2^4\right)\,\text{EllipticK}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]}{r-r2}\right)\text{Sin}\left[\frac{\lambda 1-\lambda 2}{2}\right]\text{Sin}\left[\lambda - \frac{\lambda 1}{2} - \frac{\lambda 2}{2}\right]$$

$$\text{VxyzLS} = \frac{2\,G\,\left(-r^2\,r1^5 + r1^5\,r2^2 + (r-r1)\,(r+r1)\,r2^5\right)\,\rho\,\text{Sin}[2\,\lambda-\lambda 1-\lambda 2]\,\text{Sin}[\lambda 1-\lambda 2]}{r^4\,(r-r1)\,(r+r1)\,(r-r2)\,(r+r2)}$$

$$\text{VzzxLS} = \frac{1}{r^4}\,2\,G\,\rho\,\text{Cos}\left[\lambda - \frac{\lambda 1}{2} - \frac{\lambda 2}{2}\right]\left(r1^2\left(\frac{\left(r^2-2\,r1^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]}{r+r1} - \frac{\left(r^2+2\,r1^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]}{r-r1}\right) + r2^2\left(-\frac{\left(r^2-2\,r2^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]}{r+r2} + \frac{\left(r^2+2\,r2^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]}{r-r2}\right)\right)\text{Sin}\left[\frac{\lambda 1-\lambda 2}{2}\right]$$

$$\text{VzzyLS} = \frac{1}{r^4}\,2\,G\,\rho\left(r1^2\left(\frac{\left(r^2-2\,r1^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]}{r+r1} - \frac{\left(r^2+2\,r1^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r1}{(r-r1)^2}\right]}{r-r1}\right) + r2^2\left(-\frac{\left(r^2-2\,r2^2\right)\,\text{EllipticE}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]}{r+r2} + \frac{\left(r^2+2\,r2^2\right)\,\text{EllipticK}\left[-\frac{4\,r\,r2}{(r-r2)^2}\right]}{r-r2}\right)\right)\text{Sin}\left[\frac{\lambda 1-\lambda 2}{2}\right]\text{Sin}\left[\lambda - \frac{\lambda 1}{2} - \frac{\lambda 2}{2}\right]$$